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## A Simple Problem with Integers

Language: Default ▼

**Time Limit:** 5000MS

**Memory Limit:** 131072K

**Total Submissions:** 191086

**Accepted:** 58930

**Case Time Limit:** 2000MS

## Description

You have  $N$  integers,  $A_1, A_2, \dots, A_N$ . You need to deal with two kinds of operations. One type of operation is to add some given number to each number in a given interval. The other is to ask for the sum of numbers in a given interval.

## Input

The first line contains two numbers  $N$  and  $Q$ .  $1 \leq N, Q \leq 100000$ .

The second line contains  $N$  numbers, the initial values of  $A_1, A_2, \dots, A_N$ .  $-1000000000 \leq A_i \leq 1000000000$ .

Each of the next  $Q$  lines represents an operation.

"C  $a$   $b$   $c$ " means adding  $c$  to each of  $A_a, A_{a+1}, \dots, A_b$ .  $-10000 \leq c \leq 10000$ .

"Q  $a$   $b$ " means querying the sum of  $A_a, A_{a+1}, \dots, A_b$ .

## Output

You need to answer all  $Q$  commands in order. One answer in a line.

## Sample Input

```
10 5
1 2 3 4 5 6 7 8 9 10
Q 4 4
Q 1 10
Q 2 4
C 3 6 3
Q 2 4
```

## Sample Output

```
4
55
9
15
```

## Hint

The sums may exceed the range of 32-bit integers.

## Source

POJ Monthly--2007.11.25, Yang Yi

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